

Mechanical Support in Cardiogenic Shock

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Heart disease remains the leading cause of death in both men and women among African Americans, Hispanics, and Caucasians in the United States, accounting for nearly 650,000 deaths annually.¹ Significant progress has been made over the past few decades in defining and recognizing cardiogenic shock (CS), and although there are many potential causes of this condition, acute myocardial infarction (AMI) with a large loss of functioning myocardium is the most frequent.^{2,3} Despite advances in early reperfusion and mechanical support treatment approaches, CS remains the most common cause of in-hospital mortality after AMI, with rates exceeding 50%.²

In 1912 James Herrick was the first to link observations of patients with AMI with autopsy findings of patients who had died from CS.³ His contention at the time, which was initially rejected by the medical community, was that AMI was not always immediately fatal and that efforts were needed to diagnose and treat the condition while the patient was still alive.³ It was not until the mid-to-late 20th century when selective coronary angiography became available in humans that DeWood and others reported evidence of coronary occlusion in patients suffering AMIs.⁴ These findings laid the foundational basis for developing interventional and medical treatments for AMI to intervene and support patients who were diagnosed with CS. A standard definition of CS was proposed by Binder and colleagues in 1955, which included factors such as a systolic blood pressure less than 80 mm Hg and tachycardia to greater than 100 bpm with signs of peripheral vascular collapse.⁵ At the time this definition was developed, the mortality associated with CS after AMI was still close to 100%.⁵

Over the past several decades, the general definition of CS has evolved into a severe impairment of myocardial performance that results in diminished cardiac output (CO), end-organ perfusion, and hypoxia.⁶ Despite an updated definition, a level of uncertainty in the outcomes of CS persists, with some contemporary clinical trials lacking uniformity in their definitions for identifying CS. For instance, although the SHOCK (Should We Emergently Revascularize Occluded Coronaries for Cardiogenic Shock) and Intra-Aortic Balloon Pump (IABP)-SHOCK II trials used systolic blood pressure ≤ 90 mm Hg for ≥ 30 minutes, their exact parameters for confirming end-organ hypoperfusion were diverse (Table 82.1).^{7,8} Clinically CS is recognized to be on a continuum, but the presence of hypotension refractory to volume resuscitation and features of end-organ hypoperfusion requiring pharmacologic or mechanical support are the crux of this diagnosis.⁹ AMI remains the instigator of 81% of CS events.¹⁰ Furthermore, the incidence of CS in myocardial infarction has remained relatively constant in recent decades (7%) and remains the leading cause of death in patients with AMI.⁷

HISTORY OF MECHANICAL CIRCULATORY SUPPORT

The evolution of mechanical circulatory support (MCS) dates to the early 1950s when Gibbon developed the prototype cardiopulmonary bypass (CPB) apparatus.¹¹ In the years that followed, Lillehei, Kirklin, and others applied the heart-lung machine to facilitate open heart surgery; their pioneering work and early observations led directly to the development of modern MCS systems.^{12–14} Their initial publications introduced the concept that left ventricular (LV) decompression and myocardial rest could afford enhanced cardiac recovery after the insult of open heart surgery. Clinical use of extracorporeal CPB for heart surgery became widespread in the early 1960s. Simultaneously, several groups of investigators were testing means of MCS for use outside the operating room to support patients in CS. The current modes of MCS are derivations of those originally developed.

The decision on the type of MCS used is based on the acuity of the patient's presentation, level of flow or adjunctive CO, and the overall goals of therapy. This decision often changes as the patient's clinical picture evolves. MCS devices include the intraaortic balloon pump (IABP), continuous flow pumps with or without an oxygenator, percutaneous devices, and durable continuous flow pumps. Temporary MCS devices are used in CS as a bridge to recovery (BTR), bridge-to-bridge with a long-term implantable MCS device, bridge to transplant (BTT), or bridge to decision. Some temporary MCS devices can be inserted percutaneously, whereas others require surgical implantation. Long-term MCS devices are used as a BTR, a BTT, or the definitive means of treatment for the patient's remaining lifespan—that is, destination therapy (DT). These devices have been employed for a variety of causes of CS, including AMI, inability to wean from CPB (postcardiotomy CS), acute decompensation of chronic heart failure (HF), acute myocarditis, peripartum cardiomyopathy, HF secondary to acute or chronic valvular heart disease, and congenital heart defects.

History of Aortic Counterpulsation

The concept of arterial counterpulsation was introduced in 1961 by Clauss and coworkers. It involved use of an external “ventricular” chamber that filled with blood from a catheter in the iliac artery and was subsequently compressed by a piston. Compression of the “ventricle” was synchronized to either the QRS complex of an electrocardiogram (ECG) or the impulse of a pacemaker so that a counterpulse of blood was delivered into the arterial system during diastole.¹⁵ It was demonstrated in dogs that cardiac stroke work and LV end-systolic pressures were substantially reduced with the use of a counterpulsation into the aorta. The following year, Mouloupoulos and associates adapted the model to create an IABP that could provide a similar

TABLE 82.1 Available Definitions of Cardiogenic Shock

Clinical Definition	SHOCK Trial	IABP-SHOCK II	ESC HF Guidelines
Cardiac disorder that results in both clinical and biochemical evidence of tissue hypoperfusion	Clinical criteria: SBP <90 mm Hg for >30 minutes OR support to maintain SBP >90 mm Hg AND end-organ hypoperfusion (urine output <30 mL/hr or cool extremities) Hemodynamic criteria: CI of <2.2 L·m ⁻² AND PCWP >15 mm Hg.	Clinical criteria: SBP <90 mmHg for >30 minutes OR catecholamines to maintain SBP >90 mm Hg AND clinical pulmonary congestion AND impaired organ perfusion (altered mental status, cold/clammy skin and extremities, urine output <30 mL/hr or lactate >2.0 mmol/L)	Clinical criteria: SBP <90 mm Hg with adequate volume and clinical or laboratory signs of hypoperfusion. Clinical hypoperfusion: cold extremities, oliguria, mental confusion, dizziness, narrow pulse pressure. Laboratory hypoperfusion: metabolic acidosis, elevated serum lactate, elevated serum creatinine

From van Diepen S, Katz JN, Albert NM, et al. Contemporary management of cardiogenic shock: A scientific statement from the American Heart Association. *Circulation*. 2017;136(16):e232–e268.

CI, confidence interval; PCWP, pulmonary capillary wedge pressure; SBP, systolic blood pressure.

counterpulsation without the need for blood reservoirs.¹⁶ The investigators used a balloon that was rapidly inflated and deflated with carbon dioxide during native diastole. The IABP was subsequently adapted and described for clinical use by Kantrowitz and colleagues in 1968.¹⁷ There is little difference in the modern IABP and that originally described, other than the availability of different-sized balloons (30- to 50-mL balloons), the use of helium instead of carbon dioxide, and subtle differences in the materials used to make the catheters.

History of Mechanical Assist Devices

The need for effective MCS devices with more options for greater cardiac flow support became apparent in the 1950s during the development of CPB for open heart surgery. The first attempt at isolated extracorporeal LV support was with a simple roller pump in 1962.¹⁸ Initial attempts with prolonged postoperative CPB demonstrated that the bypass circuit was damaging to both end-organ function and blood constituents after several hours of use.¹⁹ Subsequently, femoral venous-to-femoral arterial CPB was successfully used by Spencer and colleagues in four patients with postcardiotomy cardiac failure.²⁰ Simultaneous to Spencer and colleagues' work with extracorporeal systems, DeBakey designed the first intracorporeal LV assist device (LVAD), the DeBakey blood pump.²¹ A remodeled extracorporeal version was subsequently used for postcardiotomy failure in a 37-year-old woman after aortic and mitral valve replacements. The device was needed for 10 days, and the patient survived.²²

By 1972 investigators at the Texas Heart Institute had developed a pneumatically driven LVAD designed to be implanted in the abdomen.²³ This device had a blood chamber compressed by pulses of air delivered into the pump by a percutaneous driveline. Modern devices have chamber compression that is electrically powered via percutaneous drivelines. Paracorporeal, pneumatically driven devices were a parallel development. Paramount to the evolution of these devices was the sponsorship of the Artificial Heart Program of the National Heart, Lung, and Blood Institute, which was chartered in 1964.

By the 1960s continuous flow pumps, as compared with pulsatile pumps, were under development.^{24,25} Over the subsequent 15 years, continuous flow centrifugal pumps were perfected and introduced into clinical use. These pumps work on the principle of a forced, constrained vortex devised from three magnetic cones.^{26,27} They have been shown to be useful in a variety of clinical settings where short-term MCS is needed and an IABP is inadequate. Several types of small, axial-flow, or rotary pumps have also been developed, including some that allow for percutaneous deployment.^{28–41} These pumps generally contain a magnetically suspended impeller that rotates at fast rates

between 25,000 and 35,000 rpm. The latest generation of rotary pump technology comprises continuous flow devices that use fully magnetically levitated rotors that eliminate the need for seals or bearings. This technology reduces the risk of damage to blood elements and has been shown to lead to lower rates of thromboembolism.⁴²

INITIAL MANAGEMENT OF CARDIOGENIC SHOCK

The initial focus in the treatment of CS is to prevent end-organ failure through stabilization of CO and the establishment of hemodynamic stability while simultaneously diagnosing and treating the reversible causes of CS. Fig. 82.1 illustrates an algorithm for the management strategies in CS. Identification of the primary cause of CS can allow for the appropriate pharmacologic or mechanical therapies to be applied. The American Heart Association recommends that patients undergo noninvasive testing with chest x-ray, resting 12-lead ECG, and echocardiography.⁶ Suggested laboratory tests include complete blood count, electrolytes, hepatic function tests, serum creatinine, mixed venous oxygen saturation, natriuretic peptides, serial troponin levels, arterial blood gas, and lactate.⁶ The mainstay of evidence supports timely reperfusion in patients presenting with CS caused by AMI. The potential revascularization strategies can be noninvasive, with tissue fibrinogen activator or streptokinase, or invasive, with percutaneous coronary interventions (PCI) or coronary artery bypass grafting (CAGB).^{7,43,44} Vasoactive medications are often required for support in patients with CS, and the decision for which medication to initiate in CS remains challenging. The SOAP II trial (Sepsis Occurrence in Acutely Ill Patients) evaluated vasopressor selection in a subgroup of patients with CS and found that dopamine was associated with a higher rate of arrhythmias in addition to increased mortality, which may be the result of a tachycardia-induced ischemic propagation.⁴⁵ This has led to assertion of using norepinephrine as a first-line agent. Unfortunately, the methodology and clinical definitions of CS used within the SOAP II trial caused major concerns regarding the generalizability and applicability of the findings. Additional studies have suggested that epinephrine should also not be used in the initial management of CS because of its effect in increasing lactate levels, increasing oxygen consumption, lowered arrhythmogenicity thresholds, and association with higher mortality.^{46,47} Additionally, in a recent randomized clinical trial (RCT), norepinephrine was found to be superior to epinephrine in achieving hemodynamic stability without increasing lactate levels in patients in CS after AMI.⁴⁸ Escalating doses of vasopressors and inotropes in patients with CS are associated with increased mortality.⁴⁷ In the setting of refractory CS and impending or worsening end-organ failure, MCS

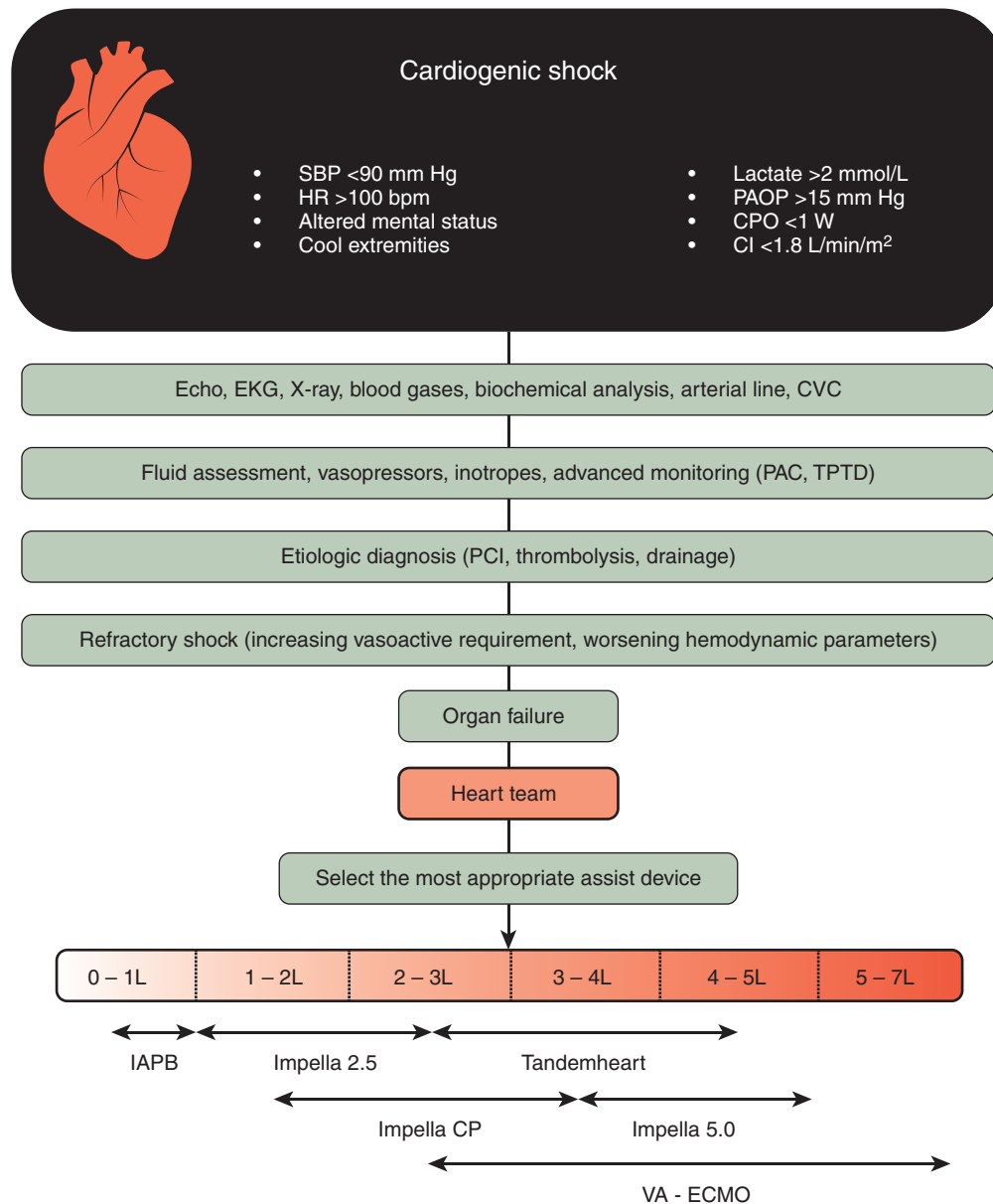


Fig. 82.1 Management schema for cardiogenic shock. (From Hajjar LA, Teboul JL. Mechanical circulatory support devices for cardiogenic shock: State of the art. *Crit Care*. 2019;23[1]:76.)

should be discussed with the multidisciplinary heart team and initiated in appropriate candidates.

The lethality of CS and limited availability of donor hearts for patients with chronic HF has been the impetus for the ongoing search for an ideal form of MCS over the past 60 years. The complexity and heterogeneity of this patient population have led to the development of a vast array of devices, none of which has proven optimal for support in all patients across the spectrum. The decision for which type of MCS device to use is based on the amount of supplemental flow required or CO that the patient requires, the duration of support required, and the patient's candidacy for cardiac transplantation (Figs. 82.2 and 82.3).⁴⁹ The biologic barriers to MCS present a constant challenge to clinicians, and the types of MCS devices available are continually evolving. Nonetheless, MCS remains an important adjunct in the treatment of CS and HF, with many devices demonstrating significant improvement in survival and quality of life. Accordingly, the American College of

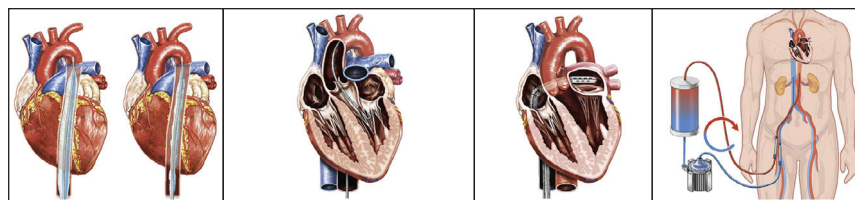
Cardiology Foundation/American Heart Association (ACCF/AHA) guidelines recommend nondurable MCS as a reasonable “bridge to recovery” or “bridge to decision” for patients with acute profound heart failure and a durable MCS for carefully selected patients with stage D heart failure and reduced ejection fraction (class IIa; level B).⁵⁰ The current indications, benefits, and limitations of MCS devices are outlined in this chapter.

CURRENT USE OF MECHANICAL CIRCULATORY SUPPORT DEVICES

Temporary Mechanical Circulatory Support

Percutaneous Insertion

Intraaortic balloon pump. The physiologic rationale for the efficacy of the IABP is that balloon deflation provides a rapid, synchronized



	IABP	Impella	TandemHeart	VA-ECMO
Cardiac flow	0.3–0.5 L/min	1–5 L/min (Impella 2.5, Impella CP, Impella 5)	2.5–5 L/min	3–7 L/min
Mechanism	Aorta	LV → AO	LA → AO	RA → AO
Maximum implant days	Weeks	7 days	14 days	Weeks
Sheath size	7–8 Fr	13–14 Fr Impella 5.0 - 21 Fr	15–17 Fr arterial 21 Fr venous	14–16 Fr arterial 18–21 Fr venous
Femoral artery size	>4 mm	Impella 2.5 & CP - 5–5.5 mm Impella 5 - 8 mm	8 mm	8 mm
Cardiac synchrony or stable rhythm	Yes	No	No	No
Afterload	↓	↓	↑	↑↑↑
MAP	↑	↑↑	↑↑	↑↑
Cardiac flow	↑	↑↑	↑↑	↑↑
Cardiac power	↑	↑↑	↑↑	↑↑
LVEDP	↓	↓↓	↓↓	↔
PCWP	↓	↓↓	↓↓	↔
LV preload	---	↓↓	↓↓	↓
Coronary perfusion	↑	↑	---	---
Myocardial oxygen demand	↓	↓↓	↔↓	↔

Fig. 82.2 Adjunctive mechanical circulatory support options and their mechanisms, implant requirements, and hemodynamic effects. AO, Aorta; IABP, intraaortic balloon pump; LA, left atrium; LV, left ventricle; LVEDP, left ventricular end-diastolic pressure; MAP, mean arterial pressure; PCWP, pulmonary capillary wedge pressure; RA, right atrium; VA-ECMO, venoarterial extracorporeal membrane oxygenation. (From Atkinson TM, Ohman EM, O'Neill WW, et al. A practical approach to mechanical circulatory support in patients undergoing percutaneous coronary intervention: An interventional perspective. *JACC Cardiovasc Interv.* 2016;9[9]:871–883.)

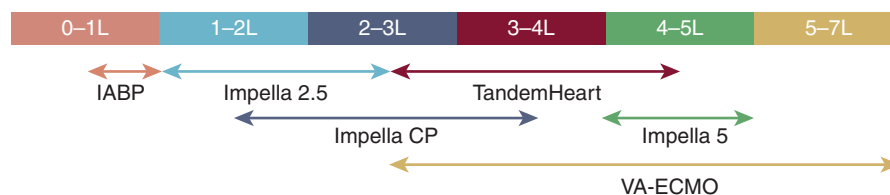


Fig. 82.3 Comparison of mechanical circulatory support devices and their impacts on cardiac flow. (See Fig. 82.2 for abbreviations.) (From Atkinson TM, Ohman EM, O'Neill WW, et al. A practical approach to mechanical circulatory support in patients undergoing percutaneous coronary intervention: An interventional perspective. *JACC Cardiovasc Interv.* 2016;9[9]:871–883.)

reduction in impedance (afterload) during isovolemic LV contraction. This is followed by a rapid, synchronized increase in aortic pressure during isovolemic LV relaxation (diastolic augmentation) caused by balloon inflation. In combination, these events achieve two important goals. First, LV systolic unloading directly reduces stroke work, which in turn reduces myocardial oxygen consumption during the cardiac cycle. Second, diastolic augmentation raises arterial blood pressure and

provides better coronary arterial perfusion during diastole, yielding increased oxygen delivery to the myocardium. The IABP does not directly move or redistribute blood flow; however, peak diastolic coronary flow velocity can be increased as much as 87% with IABP augmentation and peak diastolic flow velocity by as much as 117%.⁵¹ Since its introduction into clinical use in 1968, the IABP has remained an important adjunct to supporting patients in CS. Myocardial recovery is promoted

by the reduction of cardiac work and the simultaneous increase in myocardial oxygen supply. However, therapeutic success is dependent on the patient having a minimum degree of LV function that, in combination with IABP support, facilitates an adequate CO to sustain end-organ function. When this minimal CO is not met, alternative MCS must be considered.

The absolute indications for IABP placement include CS, uncontrolled angina pectoris, acute postinfarction ventricular septal defect (VSD), postinfarction mitral regurgitation (MR) secondary to papillary muscle rupture, and postcardiotomy left-sided HF with low CO.⁵² In these settings, IABP should be considered as a primary therapy that should not be delayed. It is important to recognize that blood pressure alone is not an adequate indication of hemodynamic or cardiac stability. Limb perfusion, renal function, mental status, and even gastrointestinal function need to be considered in the assessment of adequate resuscitation and homeostasis. Additional measurable indices include arterial (SaO₂) and mixed venous oxygen saturation, acid-base status, urine output, and body temperature. A multivariate analysis of data accrued from 391 postcardiotomy patients requiring an IABP demonstrated that epinephrine requirements greater than 0.5 µg/kg/min, a left atrial pressure greater than 15 mm Hg, urine output less than 100 mL/hr, and mixed venous oxygen saturation less than 60% correlated with mortality.⁵³ These criteria were used to help predict mortality and the need for subsequent MCS.

Other relative indications for IABP use include (1) high-risk, catheter-based interventional procedures such as left main coronary artery angioplasty, (2) after unsuccessful attempts at catheter-based intervention in patients with poorly controlled ventricular arrhythmias and concomitant poor LV function, and (3) in settings of persistent stunned, ischemic myocardium. These are circumstances in which reduction of LV systolic wall tension and oxygen consumption by the IABP might enhance myocardial recovery after intervention. The Benchmark Counterpulsation Outcomes Registry of IABP use in 22,663 patients from 250 hospitals worldwide demonstrated that CS and high-risk angioplasty were the most common indications for use of the device.⁵⁴

The optimal site of insertion of an IABP is a common femoral artery that can be accessed either percutaneously with the Seldinger technique or by surgical cutdown. Modern intraaortic balloon catheters are available for adults and children, according to the appropriate size and length for a given height and weight of the patient. Adult intraaortic balloons have a range in volume filled between 30 and 50 mL, with a standard balloon size holding 40 mL of helium. IABP catheters placed through the femoral artery are positioned so that the tip is just distal to the takeoff of the left subclavian artery in the proximal descending thoracic aorta. Optimally, the tip of the catheter should be positioned with transesophageal echocardiographic or fluoroscopic guidance.⁵⁵ To reduce the diameter of femoral cannulation, the sheathless IABP technique can be used and is our preferred method.⁵⁶

Inflation of the balloon should be timed with closure of the aortic valve (at the dicrotic notch of the aortic pressure tracing) and should be inflated to nearly occlude the descending thoracic aorta. Timing can be synchronized in one of three ways: (1) using an arterial (preferably aortic) pressure tracing in synchrony with the dicrotic notch, (2) using the descent of the R wave on a rhythm tracing, or (3) timed after a ventricular pacing spike when a pacemaker is in use.^{56–63} The optimal physiologic benefit of the IABP is significantly improved by proper timing of inflation and deflation, which can be difficult when there is an accelerated heart rate, cardiac rhythm disturbances, atrioventricular dyssynchrony, or low mean arterial pressure. Timing should be adjusted to maximize diastolic augmentation; hence, deflation should be as late as possible but just before opening of the aortic valve. If this cannot be

gauged by pressure tracing, it can be timed to the onset of the R wave on ECG tracing or with the use of M-mode echocardiography.⁶³

When femoral arterial cannulation is not desirable because of aortoiliac occlusive disease or extensive peripheral vascular disease (PVD), the subclavian artery or the ascending aorta can be used.^{64–68} With either technique, the IABP catheter is advanced antegrade down the descending thoracic aorta so that the balloon tip sits above the level of the diaphragmatic hiatus, and the most proximal end of the balloon is distal to the takeoff of the left subclavian artery. These antegrade balloons should always be placed with either fluoroscopic or echocardiographic guidance. They should be removed with open arterial repair in all cases.

IABP catheters should not be left in place after weaning because of the risk of thrombus formation and embolization. An IABP should be weaned stepwise from a rate that is equivalent to heart rate (1:1) down to a ratio of 1:4 just before removal. Balloon catheters placed via the open surgical technique should also be removed surgically. Percutaneous removal of catheters placed in the iliac artery above the inguinal ligament (most common in obese individuals) can result in significant retroperitoneal bleeding. Consideration of operative removal is warranted.

Relative contraindications to IABP use include severe atheromatous disease of the descending thoracic aorta, descending aortic dissection or aneurysm, recent descending thoracic aortic surgery, and mild to moderate aortic insufficiency. Severe aortic insufficiency is an absolute contraindication to use because diastolic augmentation cannot be accomplished, and LV end-diastolic volume and pressure are actually increased rather than decreased.

The overall complication rate of IABP use is between 5% and 10%. Major complications occur at a rate of about 3% and include severe bleeding, major limb ischemia or amputation, infection, visceral or spinal cord ischemia, and attributable IABP mortality.^{54,69} In the Benchmark registry, rates of complications were quite low; the most common complications were access site bleeding (4.3%) and limb ischemia (2.3%). The rates of amputation, stroke, visceral or spinal cord ischemia, and IABP-related mortality were all 0.1% or less.⁷⁰ Intraaortic balloon entrapment is a rare complication.^{70–72} The incidence of major vascular complications according to the STS National Database (1996–1997) and the Benchmark Registry (1997–1999) was 5.4% and 1.4%, respectively.^{55,69} Ipsilateral limb ischemia should be immediately addressed after its recognition. This usually requires removal of the IABP with replacement at another location if it is still indicated. The ischemic limb may require thrombectomy with or without revascularization and fasciotomy.^{73–79}

The SHOCK trial showed that early revascularization in CS after AMI, often facilitated by IABP use (86%), yielded a lower 6-month mortality rate (50%) than that with medical therapy alone (63%).⁷ Additional studies have shown that in patients undergoing urgent or emergent revascularization after an AMI, those supported preoperatively with an IABP had a lower operative mortality than those in whom an IABP was not used (5.3%–8.8% vs. 11.8%–28.2%).⁶⁹ These data seem to justify a strategy of aggressive IABP use to facilitate early revascularization in a postinfarction patient. The Second Angioplasty in Myocardial Infarction (PAMI-II) Trial data examined high-risk patients with AMI revascularized by PCI only and demonstrated a modest survival advantage at 6 months with the use of periprocedural IABP support.⁵⁴ When evaluating hospital mortality rates among patients undergoing CABG and/or valve surgery who received preoperative IABP or required intraoperative/postoperative IABP support, the mortality rate was significantly lower among patients supported preoperatively, as depicted in Table 82.2.^{55,58} Hence, there appears to be a survival advantage to earlier IABP support for

TABLE 82.2 Hospital Mortality (Outcome Parameter) for Patients Undergoing Cardiac Surgery Who Either Received Preoperative IABP or Intra/Postoperative IABP Support

Type of Therapy	Benchmark Registry 1997–1999 Mortality/Total Operations With IABP, N (%)	STS National Database 1996–1997 Mortality/Total Operations With IABP, N (%)	STS National Database 1996–1997 Mortality/Total Operations Without IABP, N (%)
Preoperative IABP	8.8 (329/3721)	9.5 (2487/26,077)	2.9 (10,919/378,810)
Intraoperative/ postoperative IABP	28.2 (954/3380)	23.6 (3528/14,933)	2.5 (9878/389,954)

From Christenson JT, Cohen M, Ferguson JJ 3rd, et al. Trends in intraaortic balloon counterpulsation: Complications and outcomes in cardiac surgery. *Ann Thorac Surg.* 2002;74:1086–1090.

Based on data from the Benchmark Counterpulsation Registry 1997–1999 and the STS National Database 1996–1997 compared with hospital mortality for patients who had neither preoperative nor intraoperative/postoperative IABP support.

patients with CS after AMI who undergo revascularization. In the setting of an acute VSD or acute MR after an AMI, IABP support can offer a dramatic improvement in the hemodynamic response of the patient.^{80–85} It is clear that the mortality rate of patients in CS after AMI remain high. However, these studies suggest that IABP support, combined with revascularization, portends a better prognosis than adjunctive IABP use with medical therapy alone.⁵⁴ This benefit is likely greatest in those who are revascularized and present with class 3 or 4 HF.⁸⁶ A meta-analysis of 16 studies demonstrated a significant survival benefit for the use of IABP in CS after AMI (relative risk [RR]: 0.78; confidence interval [CI]: 0.60–0.86; $P < 0.0004$).⁸⁷ However, no benefit was seen in patients with high-risk AMI that was not complicated by CS.

Conversely, a Cochrane Database meta-analysis of six RCTs reviewed the use of IABP in CS after AMI in 190 patients and found no significant improvement in in-hospital, 30-day, or 6-month all-cause mortality.^{88,89} A meta-analysis of cohort studies of IABP use in patients in CS after AMI by Sjauw and colleagues found an 18% decrease in 30-day mortality in patients treated with thrombolysis and IABP. However, in patients treated with PCI, they noted a 6% increase in mortality associated with the additional use of an IABP.⁹⁰ In the prospective IABP-SHOCK II trial, 600 patients in CS after AMI were randomized with and without placement of IABP after undergoing revascularization, primarily with PCI (>95%).⁹¹ No difference in the primary endpoint of 30-day, all-cause mortality was seen (39.7% IABP, 41.3% control). However, a 10% crossover to the IABP arm was noted. Additionally, a meta-analysis of 12 RCTs found no improvement in 30-day mortality with the use of IABP in patients with AMI, regardless of whether or not the patients suffered from CS.⁹² From this compilation of results, it may be concluded that the use of IABP in CS after AMI has little benefit in any treatment strategy that does not employ the use of early revascularization. Furthermore, improvement in the patient's hemodynamics often seen with the use of IABP does not suffice as a surrogate marker for survival in patients in CS after AMI. An IABP should be placed in the case of acute postinfarction VSD or acute MR without delay.

TandemHeart. The TandemHeart System (Cardiac Assist, Inc., Pittsburgh, PA) is an external centrifugal pump system that allows for percutaneous LV support by pumping blood from the left atrium into the femoral artery (Fig. 82.4).⁹³ Placement of the 21F venous inflow cannula is performed by accessing the femoral vein percutaneously and positioning the cannula tip in the left atrium via an atrial transeptal puncture. Outflow is established into the contralateral or ipsilateral femoral artery via a 15F or 17F cannula. The centrifugal pump is strapped to the thigh adjacent to the arterial cannula. The device contains a built-in heparin delivery system to decrease thrombin formation. An experienced team

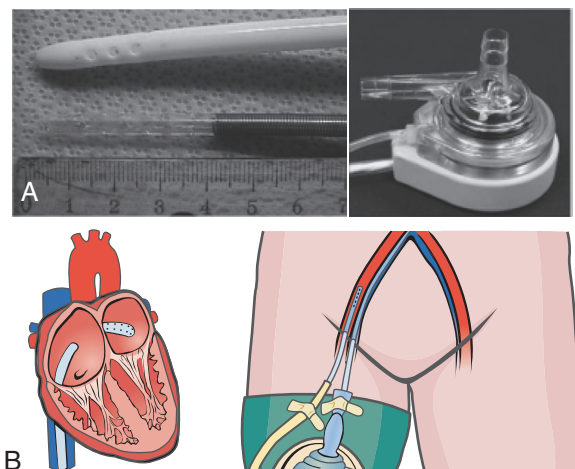


Fig. 82.4 A, Components of the TandemHeart device: 21F left atrial drainage cannula and 15–17F femoral arterial cannula (left); continuous flow centrifugal pump (right). B, Schematic demonstrating transseptal left atrial drainage and femoral access points. (From Windecker S. Percutaneous left ventricular assist devices for treatment of patients with cardiogenic shock. *Curr Opin Crit Care.* 2007;13:521–527.)

can establish MCS with TandemHeart in as little as 1 hour.⁹⁴ However, this is still longer than required for institution of extracorporeal membrane oxygenation (ECMO) support. Therefore its use may be limited in truly emergent cases and in conditions requiring cardiopulmonary resuscitation (CPR). The TandemHeart does provide more robust left-sided decompression as compared with ECMO.

Contraindications to placement of TandemHeart include severe PVD and isolated right ventricular (RV) failure. It also cannot be used in the presence of a VSD because of the potential for right-to-left shunting.⁹⁵ Right-to-left shunting is a known complication that can also occur if the transeptal cannula becomes dislodged into the right atrium (RA). Other complications include limb ischemia and thromboembolism.

TandemHeart has mostly been used as an MCS in CS.^{96–98} Support has been employed for various indications, including postcardiotomy failure,^{99–101} until corrective valvular surgery is performed,^{102–104} until recovery of fulminant myocarditis,^{105,106} or as a bridge to LVAD or BTT.^{98,107,108} It has also been used successfully for MCS during high-risk PCI.^{109–112} TandemHeart has been shown to be a more effective means of improving hemodynamic parameters in CS than IABP; however, no improvement in 30-day mortality has been demonstrated.^{113,114}

Impella. The Impella Recover axial flow pump (Abiomed, Inc., Danvers, MA) is an addition to the treatment armamentarium for

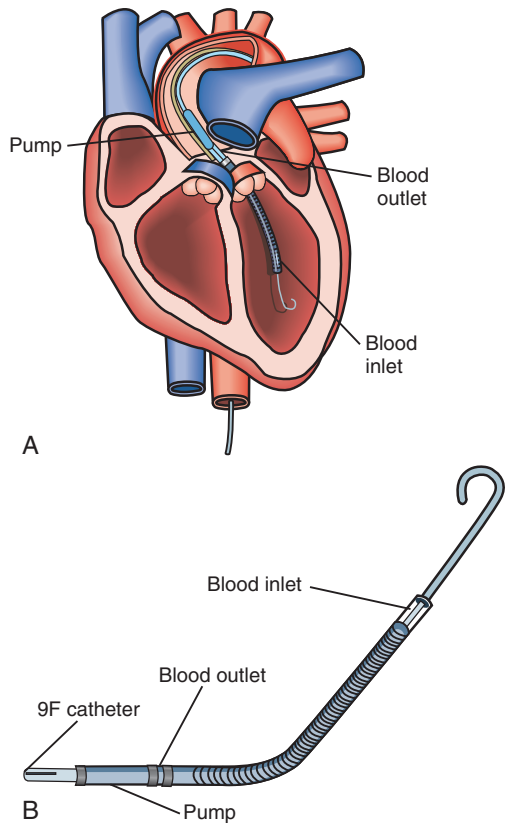


Fig. 82.5 **A**, Schematic demonstrating retrograde placement of the Impella Recover LP 2.5 device across the aortic valve. **B**, Components of the device. Blood from the ventricle enters the inlet portion of the device and is propelled by a 12F microaxial pump to the outlet portion positioned in the ascending aorta, establishing left ventricular decompression. (From Windecker S. Percutaneous left ventricular assist devices for treatment of patients with cardiogenic shock. *Curr Opin Crit Care*. 2007;13:521–527.)

acute short-term MCS (Fig. 82.5). The device is placed retrograde across the aortic valve with the tip situated in the LV. It can be placed percutaneously or surgically via the femoral artery under fluoroscopic guidance. It may also be placed surgically via an 8-mm T-graft sewn to the subclavian artery^{115,116} or directly into the ascending aorta during cardiac surgery. There are several versions of the Impella device, including the 2.5, 5.0, and CP (flow rates of 2.5, 5.0, and 5.0 L/min, respectively). Whereas the 2.5 and the CP can be placed percutaneously or surgically, the larger Impella 5.0 requires surgical placement. Contraindications for the use of Impella include patients with severe aortic stenosis and those who have had a previous mechanical aortic valve replacement. Severe PVD may make percutaneous deployment impossible or mandate a surgical cutdown for placement.

Similar to other devices, Impella has been used for MCS for several indications, including high-risk PCI^{117,118}; postcardiotomy failure^{119–121}; CS after AMI^{97,122–125}; severe allograft rejection after heart transplant^{126,127}; myocarditis^{128,129}; and as a bridge to placement of a long-term device, BTR, or BTT.⁹⁷ RCTs comparing support with Impella 2.5 with IABP in CS AMI have demonstrated improved hemodynamic and laboratory parameters (CI, mean arterial pressure, serum lactate levels) with the use of Impella.^{113,114} However, this did not translate into improved 30-day survival. In contrast, the recent PROTECT-II RCT comparing these two devices in high-risk PCI demonstrated lower rates of the two composite endpoints of major adverse events (MAEs) and major adverse cardiac and cerebral events (MACCEs = death, stroke, myocardial infarction,

and repeat revascularization).¹¹⁷ At the 3-month follow-up, the rates of both composite endpoints were lower in the Impella group than in the IABP group (MAE, 37% vs. 49%, $P = 0.014$; MACCE, 22% vs. 31%, $P = 0.034$) at 3 months.

In addition to the Impella devices for LV support, the Impella RP is available for RV support. The Impella RP is a 22F percutaneous, microaxial pump mounted on an 11F catheter. The catheter pump is advanced antegrade through the femoral vein under fluoroscopic guidance and positioned across the tricuspid and pulmonic valves.¹³⁰ The pump inflow aspirates blood from the inferior vena cava and ejects it into the pulmonary artery at a rate of up to 4.0 liters per minute. The intended use for the Impella RP is 14 days; however, if there is echocardiographic and hemodynamic evidence of improved RV function, the device can be weaned earlier. During pump operation, an activated clotting time of 160–180 seconds is required. The RECOVER RIGHT study was a prospective, multicenter trial that demonstrated improved hemodynamics and a 73% 30-day survival in patients with RV failure after AMI.¹³¹

Protek duo. The tandem heart Protek Duo is a device used for isolated RV support accessed via the right internal jugular vein. The device contains two lumens within the 29F or 31F cannula.¹³⁰ One serves as the inflow cannula, which is positioned across the superior vena cava and RA, where it removes blood and delivers it to a centrifugal pump. The second lumen is the outflow cannula, which has a multifenestrated tip to deliver blood into the main PA. This system effectively bypasses the RV. The Protek Duo can flow between 2 and 4 liters per minute.¹³² The Protek Duo has been reported for use after AMI with RV infarcts, myocarditis, posttransplant RV failure, postcardiotomy RV failure and post-LVAD RV failure.¹³³

Extracorporeal membrane oxygenation. Since the 1970s, ECMO has been used in the adult population for short-term MCS when CS is complicated by concomitant pulmonary insufficiency. It provides full cardiopulmonary support to allow reversal of the systemic malperfusion that occurs in CS until definitive surgical correction is performed. It can also serve as a BTR, a BTT, or a bridge-to-bridge to longer-term MCS (i.e., VAD).^{134–136} It may be employed for a period of 1–3 days in cases when the neurologic status of a patient is unclear, but longer-term support is generally not appropriate until this status is clarified.

The ECMO circuit consists of inflow and outflow cannulas, a continuous flow centrifugal pump, a hollow-fiber oxygenator, and a heat exchanger. Historically, roller pumps and centrifugal pumps have both been used. Roller pumps remain in widespread use for CPB during cardiac surgery; applications outside the operating room have been virtually abandoned because of higher rates of circuit disruption, particle emboli, and hemolysis.¹³⁷ Bio-Medicus Biopump (Medtronic Corp., Minneapolis, MN) and CentriMag (Levitronix LLC, Waltham, MA) are two centrifugal pumps commonly used in ECMO circuits. The Bio-Medicus Biopump generates a constrained vortex within an acrylic shell that houses concentric magnetic cones. The cones rotate as a magnetic rotary motor spins adjacent to the base of the cones and can generate very high flows with less trauma to blood cells than roller pumps.^{137–141} The cannulation strategy most often used in acute CS is peripherally via the femoral vessels with either the percutaneous Seldinger or surgical technique. Alternatively, central cannulation via the RA and aorta is used to provide temporary MCS in the case of postcardiotomy failure or in patients who require further ventricular decompression with LV vents in lieu of left thoracotomy drainage approaches or additional percutaneous device therapy. When faced with acute refractory CS, ECMO can facilitate patient stabilization for subsequent transport to a tertiary medical center for VAD evaluation. One particular ECMO system is the CardioHelp (Maquet Cardiopulmonary AG, Hechingen, Germany). It is a miniaturized system that combines the

pump and oxygenator in a single unit. Considering the economical size, it is ideal for patients who require urgent interhospital transport.¹⁴²

Disadvantages in using the peripheral cardiopulmonary support or ECMO include the greater potential for ipsilateral limb complications, higher rates of hemolysis, the requirement for anticoagulation to prevent thrombosis of the oxygenator and circuit, and failure to adequately decompress the left ventricle.^{143–149} Inadequate LV decompression with peripheral cardiopulmonary support/ECMO systems may be the mechanism responsible for some treatment failures. Regardless of the etiology of CS, a rested ventricle (i.e., decompressed) has a better chance of recovery than a distended ventricle. Lower extremity ischemia may be resolved or prevented by placing an additional perfusion cannula (8–14F) down the ipsilateral superficial femoral artery.¹⁵⁰ Alternatively, limb ischemia may be prevented with the use of an 8-mm T-graft sutured to the side of the femoral artery.¹⁵¹ However, this increases the complexity of instituting ECMO and can be impractical in cases of CS with ongoing CPR where efficiency is key.

The use of ECMO in the adult population for reasons other than primary cardiac failure with secondary pulmonary insufficiency has limited advantages over conventional therapies.^{151,152} However, a substantial subset of patients who present with CS and are initially resuscitated with cardiopulmonary support/ECMO survive to revascularization, transplantation, or recovery, with survival rates as high as 75%.^{153–164} ECMO used as a bridge to VAD placement for profound CS can yield survival rates higher than 40%.¹⁶⁵ This strategy is pragmatic and offers immediate end-organ support while a subsequent definitive treatment plan can be designed. In an outcome study by Combes and colleagues, early predictors of death on ECMO support include hepatic failure, renal failure, or placement of ECMO while undergoing CPR. Of the greater than 40% who did survive to discharge, many reported continued physical and social problems, including the ability to return to work.¹⁶⁶ This underscores the importance of long-term follow-up and the need for continued psychosocial support of these patients.

Improvements in pump and oxygenator technology have made ECMO a relatively simple means to establish MCS, allowing reversal of end-organ malperfusion that inevitably accompanies CS not promptly corrected with medical therapy alone. However, complications of bleeding, infection, stroke, and limb ischemia increase with time on ECMO support.¹⁶⁷ As with any form of MCS, the plan for separation from support or transition to a more permanent means begins upon institution of ECMO. Once end-organ malperfusion has been corrected, the feasibility of ECMO weaning is determined on a daily basis. Weaning trials are guided based on the patient's hemodynamic parameters, oxygenation, and ventilation. Pulmonary and circulatory support can be weaned independently or simultaneously. Pulmonary weaning consists of increasing support from mechanical ventilation while the amount of oxygenation and ventilation provided by ECMO is reduced. Circulatory weaning is done by gradually decreasing the amount of pump flow in a stepwise fashion while assessing hemodynamic and echocardiographic parameters to evaluate the cardiac response to ventricular loading that occurs with decreasing pump flow. Once it is determined that the patient can maintain adequate perfusion and pulmonary function, a plan is made to separate the patient from ECMO and remove the cannulas. The cannulas are most often removed in the operating room with direct vascular repair of the femoral artery. If the patient cannot be successfully weaned from ECMO, the plan for conversion to VAD or transplant is undertaken. This period of support, weaning, and determining the long-term strategy is done with the collaboration of a multidisciplinary heart team, including surgeons, cardiologists, and critical care providers.

Temporary Surgically Inserted Devices

Short-term MCS using continuous flow pumps for CS is a relatively simple means of establishing immediate and complete MCS, requiring

no additional equipment other than that needed for standard CPB support during cardiac surgery. As previously discussed, some MCS pumps can be used with percutaneous cannulation for ECMO support. These same pumps may be placed by surgical insertion. The most common indication for temporary MCS where surgical insertion is used is for postcardiotomy CS. This can be achieved by conversion from the CPB circuit used during the primary cardiac operation to a temporary VAD system, using the existing cannulas in the RA and aorta. Temporary VAD support may be required for several days to allow myocardial recovery. If the decision is made to leave the chest open, the cannulas can be brought out through the lower portion of the sternotomy incision and covered with a sealed temporary dressing. Alternatively, they can be brought out through separate incisions inferior to the sternotomy, allowing the sternum to be closed temporarily. The patient can then be brought back to the operating room within several days to explant the cannulas once cardiac recovery has occurred or converted to a long-term LVAD. Under the most recent United Network for Organ Sharing (UNOS) cardiac allocation policy change on October 18, 2018, the priority of patients on nondischARGEABLE, surgically implanted MCS has increased. As such, these devices are frequently considered in patients who are potential candidates for BTT.¹⁶⁸ Temporary, surgically implanted MCS can be withdrawn in cases where severe multiorgan failure ensues despite therapy or conversion to long-term MCS or BTT is not appropriate (advanced age, lack of appropriate psychosocial support). However, it is best to determine candidacy for definitive treatment with the multidisciplinary heart team initially and discuss the goals of treatment before performing any high-risk surgical case.

CentriMag ventricular assist. The CentriMag system uses fully magnetically levitated technology to provide MCS in a fashion similar to the Bio-Medicus Biopump. The CentriMag system has many advantages that make it attractive for temporary MCS in the acute setting.¹⁶⁹ These advantages include ease of implantation, direct outflow cannulation of the ventricle for improved decompression, and less damage to blood elements compared with traditional devices such as the Bio-Medicus pump. It has been used effectively for univentricular or biventricular support in the setting of postcardiotomy failure as a bridge to decision, BTR, BTT, or bridge to a long-term MCS device. For patients with concomitant pulmonary insufficiency, an oxygenator may be spliced into the circuit, effectively converting it to an ECMO system.

Durable Ventricular Assist Devices

Continuous Flow Left Ventricular Assist Devices

In the continued search for smaller, implantable LVADs with increased durability, many advances in technology have led to an array of continuous flow pumps, each with unique features. These axial or centrifugal flow pumps are smaller and more durable than pulsatile devices, making them well suited for long-term support as a BTT, BTR, or DT.^{170–172} They are placed within the pericardium or in a preperitoneal pocket, with only an external driveline exposed to provide power and device control. These devices are relatively costly, provide isolated LV support, and require specialized training to implant, although they provide significantly improved survival and quality-of-life outcomes over patients with refractory heart failure treated with optimal medical therapy. In the setting of limited donor organs, these devices continue to be a significant treatment approach for patients with HF.

HeartMate II (HMII). HeartMate II (Abbott Inc., Chicago, IL) is an implantable axial flow LVAD that has been used as a BTT and DT.^{171–174} The device is implanted in a preperitoneal pocket with the inflow cannula situated in the LV apex. Outflow is via a graft anastomosed to the ascending aorta. The device was approved for DT by the Food and Drug Administration (FDA) in 2010. A prospective postapproval study confirmed the efficacy of the device and demonstrated a reduction in

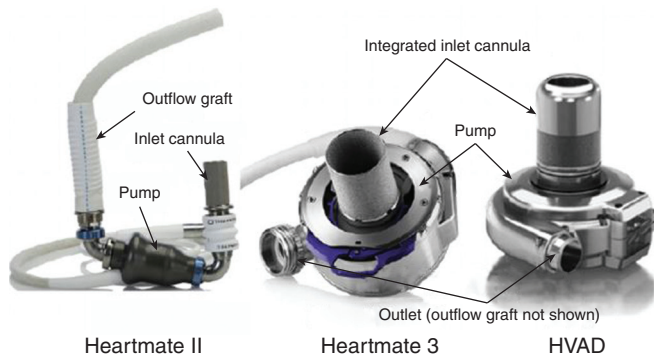


Fig. 82.6 Implantable continuous flow ventricular assist devices currently in common clinical use. (From Stulak JM, Abou El Ela A, Pagan FD. Implantation of a durable left ventricular assist device: How I teach it. *Ann Thorac Surg*. 2017;103[6]:1687–1692.)

the median length of stay by 6 days as compared with the pivotal DT trial.¹⁷⁵ Results of the ROADMAP trial demonstrated that limited HF patients treated with the HeartMate II LVAD had significant improvement in the primary outcome composite of survival and improvement in a 6-minute walk test at 1 year compared with similar patients treated with optimal medical therapy.¹⁷⁶

Jarvik 2000. Jarvik 2000 (Jarvik Heart, Inc., New York, NY) is an implantable axial flow pump used as a BTT and for DT.^{177,178} It is unique in that the pump itself resides within the LV cavity, providing outflow via the LV apex to a graft anastomosed to the ascending or descending thoracic aorta. It is a direct current (DC) battery-powered device, using alternating current (AC) power only to recharge the batteries.

HeartWare ventricular assist device (HVAD). The HVAD (HeartWare, Inc., Framingham, MA) is a centrifugal pump that is implanted within the pericardial space. The pump is attached directly to the LV apex where inflow is provided. Outflow is via a graft sewn to the ascending aorta. Based on results from the ADVANCE clinical trial, the HVAD received FDA approval as a BTT device.¹⁷⁹ Additionally, the ENDURANCE trial found centrifugal flow pumps (HVAD) to be non-inferior to axial flow pumps (Heartmate II), with higher rates of strokes (29.7% vs. 12.1%, $P < 0.001$) reported in the HVAD group.¹⁸⁰ The ENDURANCE supplemental study subsequently confirmed the associations between poorly controlled hypertension and strokes in LVAD patients and thus the importance of blood pressure management in the LVAD population.¹⁸¹

Heartmate III. The Heartmate III (Abbott, Inc., Chicago, IL) device is a centrifugal flow pump with a bearingless, fully magnetic levitated motor with wide blood-flow passages and no mechanical bearings (Fig. 82.6).¹⁸² Additionally, the Heartmate III is programmed to facilitate rapid changes in rotor speed to create an intrinsic artificial pulse that can potentially reduce stasis and thrombus within the pump.¹⁸³ The MOMENTUM 3 trial compared the Heartmate II axial flow pump with the Heartmate III centrifugal flow pump. At 2-year follow-up, the investigators found that the Heartmate III was superior with fewer disabling strokes, reoperations for pump thrombosis, and major bleeding events when compared with the Heartmate II.¹⁸⁴

TREATMENT OF CARDIOGENIC SHOCK: AN ALGORITHM FOR MECHANICAL CIRCULATORY SUPPORT

The hallmarks of CS are low CO, hypotension, peripheral vasoconstriction, cold extremities, poor urine output, and altered mental status. As

the pathophysiologic state progresses, pulmonary insufficiency and pulmonary edema ensue. Extrinsic causes of CS most commonly manifest as circulatory collapse secondary to pericardial tamponade. Acute tamponade is easily diagnosed by echocardiography and requires surgical or percutaneous evacuation and subsequent treatment of that which caused the tamponade (e.g., traumatic injury, aortic dissection, ruptured aneurysm). Extrinsic causes of CS usually require immediate surgical intervention but rarely necessitate mechanical assistance. However, intrinsic causes of CS can be refractory to both medical and surgical therapies and may require MCS. Intrinsic causes can be divided into four pathophysiologic classifications: (1) acute valvular insufficiency, (2) AMI, (3) acute myocarditis, and (4) postcardiotomy cardiac failure. Irrespective of the etiology of CS, the approach toward the initial management of patients should be fairly uniform. Irrespective of the etiology of CS, the approach toward the initial management of patients should be fairly uniform. A suggested management algorithm is outlined in Fig. 82.7.

First, insertion of a pulmonary arterial catheter and echocardiography should be done to help formulate a differential diagnosis. Severe valvular insufficiency or a VSD can usually be effectively excluded at this juncture. If severe aortic insufficiency is present, chronotropic control (heart rate 80–100 beats/min) and afterload reduction with inotropic support should be the initial maneuvers. An IABP is contraindicated because aortic regurgitation will worsen; these patients should be prepared for immediate aortic valve replacement. For patients with acute MR or a VSD, an IABP should be placed immediately in conjunction with inotropic support and afterload reduction. Surgical intervention should proceed emergently. Cardiac catheterization is pursued preoperatively to identify coronary lesions that would require CABG or if PCI is more appropriate.

Acute fulminant myocarditis usually presents in a previously healthy individual with no history of cardiac disease. Patients with presumed myocarditis who do not stabilize after the insertion of an IABP and concomitant inotropic infusion should be diverted to further temporary VAD support expeditiously. A remarkable percentage of these patients will recover if adequately supported during the acute phase of this disease. Short-term to intermediate-term VADs are optimal in these patients because of the ease of their insertion and removal and the anticipation for relatively short-term recovery. Giant cell myocarditis is one exception to this rule, because most patients with this diagnosis will require transplantation or durable MCS.^{185–187}

AMI requires pharmacologic support along with emergent cardiac catheterization and PCI as indicated. MCS can be rapidly instituted with IABP placement and should be considered in conjunction with PCI. If a mechanical complication (i.e., severe MR or VSD) has occurred, an IABP should be placed without delay. Urgent surgical intervention is usually required for these mechanical complications. The culprit vessel resulting in the AMI is treated immediately with PCI. The number of diseased arteries remaining and the clinical status usually determine subsequent allocation to further PCI or CABG. This can be accomplished several days later, once the myocardium is recovered and the shock state has resolved. Patients who remain in CS after AMI despite PCI, IABP, and inotropic support should be considered for VAD support.

Postcardiotomy CS should be managed intraoperatively with an initial trial of IABP and inotropic support. If there is persistent shock or an inability to be weaned from CPB, VAD implantation is the next therapeutic step, provided a meaningful recovery is predictable or a plan for transplantation or permanent therapy can be clarified. The general recommendation for implantation of a VAD for postcardiotomy failure is within 1 hour of the first failed attempt to wean from CPB.

The mode of MCS used for CS is determined by a number of factors. First is the degree of pulmonary insufficiency. If there is pulmonary

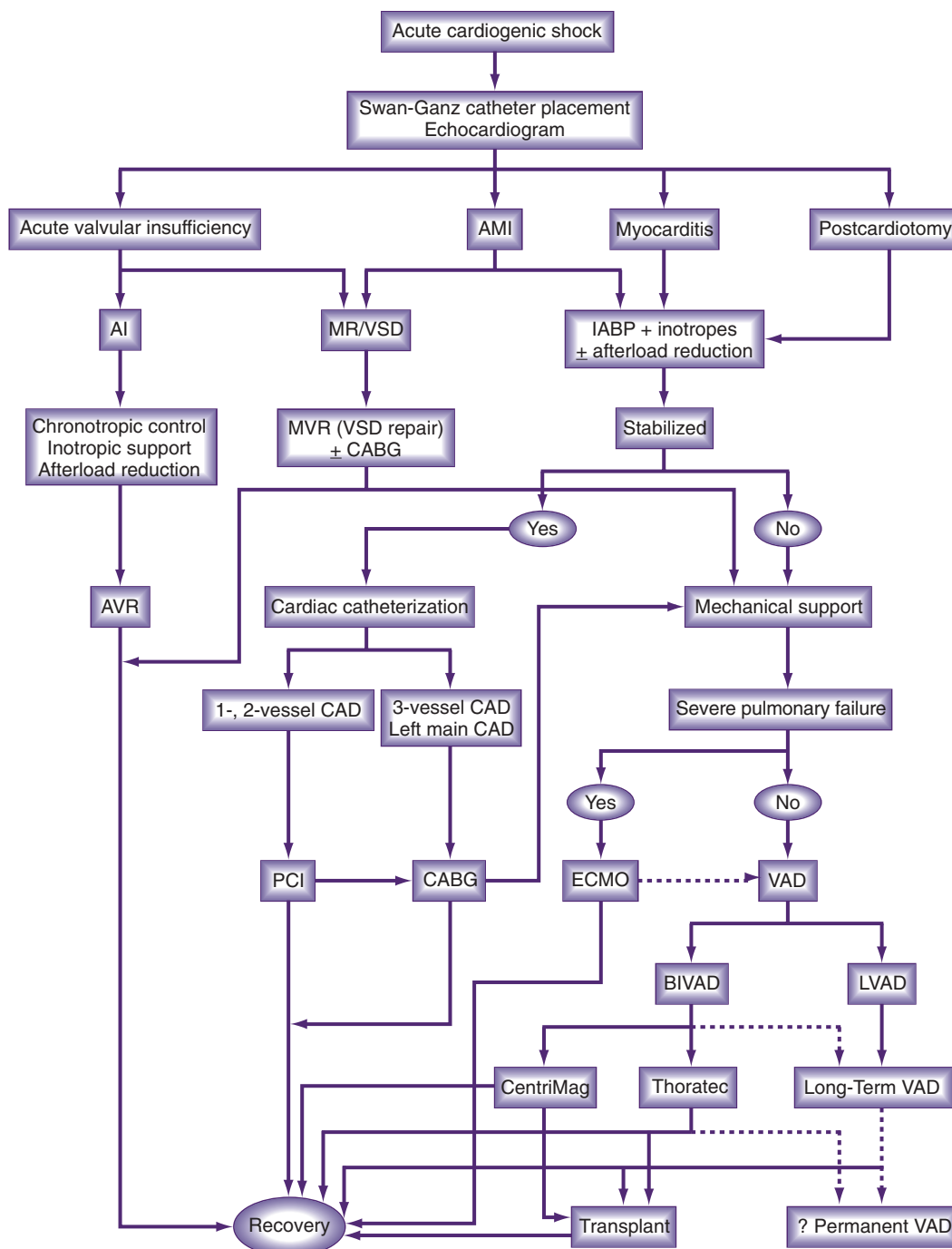


Fig. 82.7 Algorithm for the management of acute cardiogenic shock.

failure with a very large alveolar-to-arterial oxygen gradient on maximal ventilatory support, ECMO support is indicated. A small percentage of ECMO patients in this setting will recover, some will require VAD placement as a BTT, and fewer still will bridge to VAD and then to recovery. If the degree of pulmonary insufficiency is limited to pulmonary edema that is likely to recover with adequate CO, patients should undergo VAD placement directly. The choice of VAD in this situation is also dependent on several factors, including the predicted need for short- or longer-term support, the need for univentricular versus biventricular support, the chance of ventricular recovery, the institutional experience with different devices, device availability, and the relative risks of anticoagulation.¹⁸⁷

The Levitronix CentriMag system is an attractive option for postcardiotomy cardiac failure in those patients predicted to recover within days to a week of surgery for cases when neurologic function is not known or is markedly compromised but may need a BTR or bridge to a durable device or BTT once stabilized. Both are easy to insert. Therefore in cases of profound CS when operative brevity is paramount, these devices may be beneficial. Additionally, the CentriMag devices allow for arterial cannulation via an inflow graft sewn onto the aorta, which may make conversion to other long-term VAD systems technically easier.

Initial placement of continuous flow VADs (e.g., HeartMate II, Heartmate III or HeartWare HVAD) for MCS in patients with CS is

generally not indicated. These devices may be used as a second bridge (“bridge-to-bridge”) toward recovery, BTT, or DT. Typically the only patients with CS who are candidates for direct, durable LVAD placement are those who are more stable without profound end-organ failure, can sustain longer operative times, and are unlikely to achieve myocardial recovery.¹⁸⁸

OUTCOMES OF CARIOGENIC SHOCK WITH MCS SUPPORT

As recently as 2017, there were a reported 25,145 adult patients who had undergone MCS implant, with the majority (76.3%) receiving durable, continuous flow LVADs and a minority (2.7%) receiving biventricular support.¹⁸⁹ Although the burden of adverse events remains high for these patients, overall survival at 1 year is >80% and approximately 70% at 2 years.¹⁹⁰ The most commonly encountered adverse events in patients on durable LVAD support, listed in decreasing incidence of hazard, are neurologic dysfunction, multisystem organ failure, infection, device malfunction, and right heart failure.

In adult patients with refractory CS requiring ECMO support, survival to hospital discharge currently ranges from 42% to 66.7%,¹⁹¹ although patients with chronic renal failure, pre-ECMO end-organ failure, pre-ECMO cardiac arrest, congenital heart disease, lower pulse pressures, and lower serum bicarbonate were at higher risk for mortality.¹⁹² Patients with postcardiotomy CS requiring ECMO have traditionally been poor, with in-hospital mortality rates between 59% and 84%; however, in the modern era, successful weaning from postcardiotomy ECMO has been reported in up to 63.3% of patients, with survival to discharge near 25%.¹⁹³

The evidence surrounding the use of Impella devices is continuing to evolve. A propensity-matched study compared CS patients after an AMI who were supported with an Impella 2.5, Impella CP, IABP or only medical therapy found Impella to be superior to IABP and medical therapy.¹⁹⁴ Furthermore, the IMPRESS (Impella versus IABP Reduces Mortality in STEMI Patients Treated with Primary PCI in Severe Cardiogenic Shock) in Severe Shock trial demonstrated that patients randomized to an Impella CP had similar survival outcomes at 2 years when compared with IABP-supported patients.¹⁹⁵

HEART TEAM APPROACH TO CS REQUIRING MCS

In the setting of CS, the multidisciplinary heart team, which includes HF cardiologists, interventional cardiologists, HF surgeons, and intensivists, helps evaluate each individual patient and develop a care plan that best suits the individual patient’s needs. The hallmarks of a successful heart team approach for CS requiring MCS include early activation, rapid initiation of revascularization and MCS when appropriate, hemodynamic-guided management, and strict protocol adherence.¹⁹⁶ The implementation of a dedicated heart team approach in patients with CS requiring MCS has been shown to improve 30-day survival from 47% to 77%.¹⁹⁷ An example of the heart team approach for CS patients is illustrated in Fig. 82.8.¹⁹⁸ Key aspects include an immediate heart team activation call once CS is suspected. If the patient is within the hospital, they are immediately transferred to the cardiac catheterization laboratory or the cardiac intensive care unit. If the patient is outside of the hospital system, they are emergently transferred by air if available. If the patient is not actively in an arrest situation, hemodynamic, angiographic, and laboratory data are collected and then the decision to pursue MCS is evaluated by the heart team. If an arrest situation is evolving, the

heart team will discuss the patient’s appropriateness for either ECMO or an alternative percutaneous MCS approach.

CONCLUSION

CS remains a lethal problem with a mortality rate as high as 75%.^{3,187,188} Patients who cannot be stabilized with inotropic support and an IABP should be considered for MCS with a VAD. The use of MCS for acute CS has facilitated impressive improvements in survival for certain disease cohorts, such as those with acute myocarditis, with survival rates over 70%.¹⁵⁴ VADs have had less remarkable an impact on patients with postcardiotomy shock or CS after AMI, but results in these patient populations are continuing to improve.¹⁶¹ Inherent to achieving better results is our understanding that patients who present with CS typically have significant underlying comorbidities with multiple-system organ dysfunction and marked derangements in both coagulation and inflammatory mediators that complicate management. They need to be approached by an integrated multidisciplinary team, including cardiologists, cardiac surgeons, anesthesiologists, critical care specialists, and experienced nursing staff, to implement efficient and decisive treatment plans. These integrated systems offer the greatest chance for salvage.^{197,198} The future holds exciting promise with the continued expansion of available devices, both temporary and durable, which will no doubt continue to improve outcomes in this challenging population.

KEY POINTS

- The leading cause of death among hospitalized patients with AMI continues to be CS. Mortality remains high at approximately 40%.
- The hallmarks of CS and low CO are hypotension, peripheral vasoconstriction, cold extremities, poor urine output, and altered mental status.
- Intrinsic causes of CS can be divided into four pathophysiologic classifications: (1) acute valvular insufficiency, (2) AMI, (3) acute myocarditis, and (4) postcardiotomy cardiac failure.
- Echocardiography should be done to help formulate a differential diagnosis, and insertion of a pulmonary artery balloon catheter should be considered.
- CS with AMI requires immediate cardiac catheterization. Concomitant IABP placement or other temporary VAD support may be considered in addition to pharmacologic support for patients with refractory CS.
- Pioneering surgeons recognized by the 1960s that LV decompression and myocardial rest could afford enhanced cardiac recovery.
- The physiologic rationale for the efficacy of the IABP includes (1) LV systolic unloading directly reduces stroke work, which in turn reduces myocardial oxygen consumption during the cardiac cycle, and (2) diastolic augmentation, which raises arterial blood pressure and provides better coronary arterial perfusion during diastole, yielding increased oxygen delivery to the myocardium.
- Relative contraindications to IABP use include severe atheromatous and atherosclerotic descending thoracic aorta, descending aortic aneurysm, recent descending thoracic aortic surgery, and mild to moderate aortic insufficiency.
- Short-term cardiopulmonary support for CS is an important adjunctive therapy. It is a relatively simple means of establishing immediate and complete circulatory support, requiring no additional equipment other than that needed for standard CPB support during cardiac surgery.
- Initial placement of durable VADs for MCS in patients with acute CS is generally not indicated. These devices are best suited as a bridge-to-bridge once the shock state has resolved, often with a temporary MCS device.

 References for this chapter can be found at expertconsult.com.

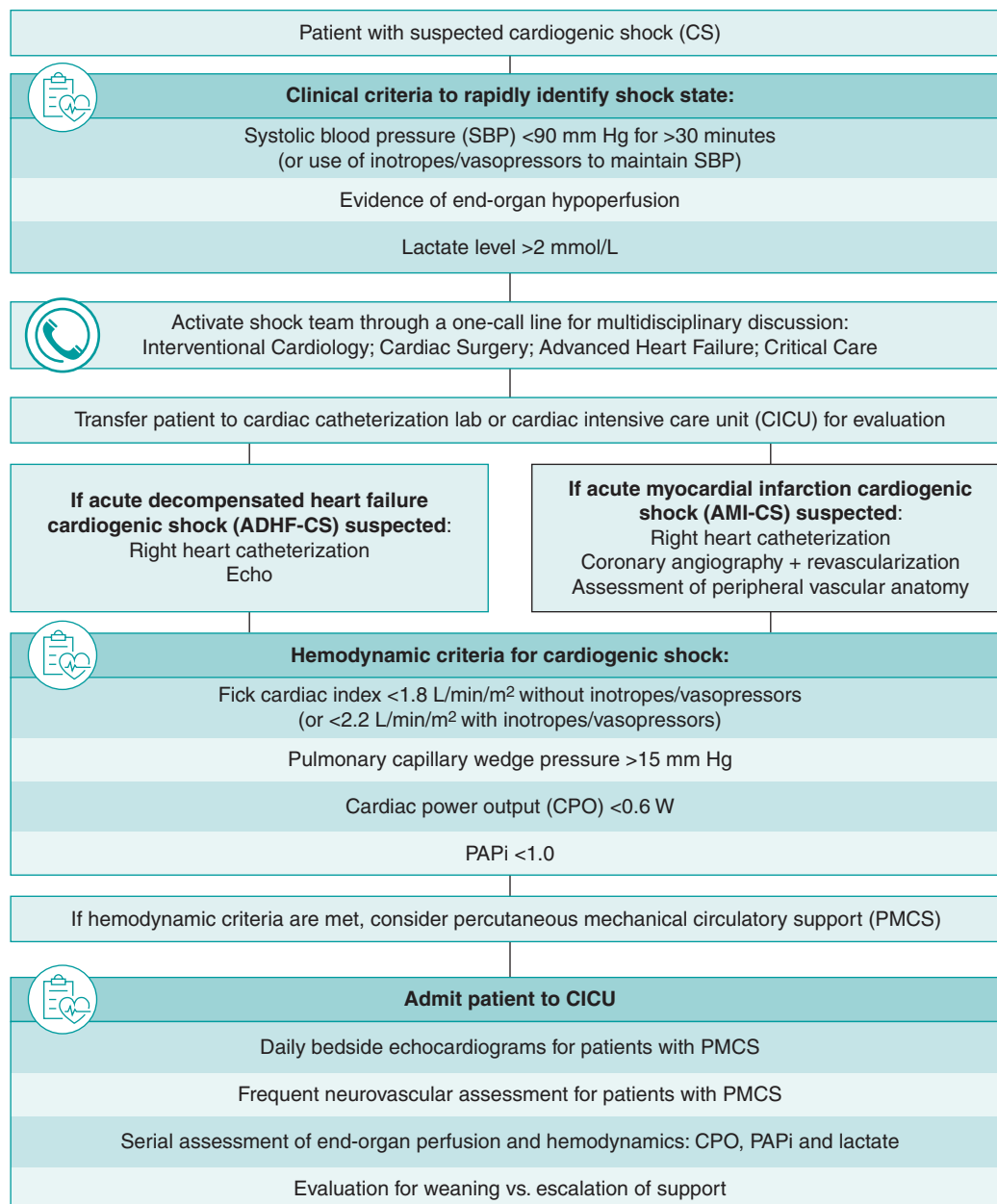


Fig. 82.8 Algorithm for the management from the integration of a multidisciplinary heart team in the management of cardiogenic shock. (From Tehrani BN, Truesdell AG, Sherwood MW, et al. Standardized team-based care for cardiogenic shock. *J Am Coll Cardiol*. 2019;73[13]:1659–1669.)

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- in lactate levels, suggesting improved end-organ perfusion. However, the mortality rates of Impella 2.5–supported CS patients remained high at 64.2%. The major complications that were most frequently seen were bleeding, hemolysis, and pericardial tamponade. Age above 65 years and lactate >3.8 mmol/L at admission were identified as predictors of 30-day mortality.
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